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SECTOR AT A GLANCE | 10

Anti-microbial Resistance (AMR)

Health Sector in India

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The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Anti-microbial Resistance (AMR) also referred to as the “Other” Pandemic is a major public health concern globally and is projected to kill ten million people annually by the year 2050 as against 0.7 million per year currently.¹ However, the problem is nowhere as stark as in India where the burden of infectious diseases is high and consumption of antibiotics is huge and unregulated.²

India exhibits some of the highest antibiotic resistance rates, especially among common infection-causing bacteria in both community and healthcare settings. Resistance surpasses 70% for broad-spectrum antibiotics like fluoroquinolones and third-generation cephalosporin in *Acinetobacter baumannii*, *Escherichia coli* (E.coli), and *Klebsiella pneumoniae*, with rates exceeding 50% in *Pseudomonas aeruginosa*.^{3,4}

Antimicrobial Resistance: A Global Threat



The global emergence and spread of drug-resistant pathogens, leading to AMR, threatens the ability to combat common infections. Rapidly spreading multi- and pan-resistant bacteria are becoming untreatable with existing antimicrobial medicines.

The clinical pipeline for new antimicrobials is insufficient, with only a handful of innovative antibiotics in development according to the **2019 World Health Organization (WHO) AWaRe classification of antibiotics for evaluation and monitoring of use**⁵. Development of new antibacterials, especially for carbapenem-resistant gram-negative infections is imperative. A failure to change antibiotic usage patterns could render new antibiotics ineffective. The economic impact of AMR is significant, affecting productivity and healthcare costs. Without effective prevention and treatment tools, more people will experience treatment failures and fatalities, making medical procedures riskier.⁶

Antimicrobial Resistance (AMR) is recognised as a complex problem and addressing it requires countries to make joint efforts across various disciplines. Considering the complex nature of the AMR problem, no individual country can address this major public health problem independently. It requires countries to conduct collaborative research across multiple disciplines to come up with comprehensive and creative solutions to overcome AMR.²

Environment plays a pivotal role in the development, transmission, and dissemination of AMR. Bacteria are present everywhere, including the soil, water and air and it is imperative to acknowledge the interconnectedness and indivisibility of humans, animals, plants and the environment at global, regional and local levels. It calls for collaboration across all sectors, involving various stakeholders and institutions.⁷

Prevention lies at the heart of the measures necessary to curb the emergence of AMR, and the environment is an integral component of this solution. Recognising the environmental dimension of AMR is essential for creating effective strategies that safeguard public health and the well-being of ecosystems.

1 Singhal. Antimicrobial Resistance: The 'Other' Pandemic. Indian J Pediatrics.

2 Department of Biotechnology. (2017). Scoping Report on Antimicrobial Resistance in India.

3 Indian Council of Medical Research. AMRSN Initiative. Retrieved 25th October 2023.

4 Mogasale, A descriptive analysis of antimicrobial resistance patterns of WHO priority pathogens isolated in children from a tertiary care hospital in India. Sci Rep

5 World Health Organization. (2019). AWaRe Classification Database.

6 World Health Organisation. Antimicrobial Resistance. Retrieved 20 September 2023.

7 UN Environment Programme. Antimicrobial Resistance, A Global Threat. Retrieved 20 September 2023

8 World Bank. Drug-Resistant Infections Could Cause Global Economic Damage. Retrieved 20 September 2023

9 World Health Organisation. (2022). Global Tuberculosis Report.

10 National Centre for Disease Control. (2017). Containment of AMR, NCDC.

Rising Focus on Antimicrobial Resistance

- WHO has declared AMR as one of the **top ten global public health threats** faced by humanity.⁶
- In the year 2019, **12.7 Lakh deaths** were directly associated with drug-resistant infection globally.⁷
- According to World Bank Group, Low-income countries could face a **5% loss of the GDP** owing to AMR. An additional 2.8 Crore people (of which 2.6 Crore from low-income countries) could be pushed into poverty by 2050 owing to AMR.⁸
- AMR could increase global healthcare costs ranging from **USD 300 billion to USD 1 trillion** per year by 2050.⁸
- Implications of AMR on output and trade in livestock and livestock products are particularly susceptible to adverse effects. A WHO report mentions that by 2050, there could be a decline in global livestock production ranging from **2.6% to 7.5% annually**, if the AMR continues to spread unchecked.⁸
- In 2021, **India held the largest share** of incident cases of Multidrug-resistant or rifampicin-resistant tuberculosis (MDR/RR-TB) globally, accounting for approximately **26% of the total global cases**.⁹
- The prevalence of bacterial infections is significant in India, with **approximately 4,10,000 children** under the age of five succumbing to pneumonia each year. Pneumonia accounts for **25%** of all child fatalities in the country.¹⁰

“Economic Costs of AMR may be as severe as during the 2008 Financial Crisis”

– World Bank⁸

The **Ministry of Health and Family Welfare (MoHFW)** addresses AMR in India. Within the ministry, various departments and agencies are involved in developing policies, guidelines, and strategies to combat AMR. Additionally, the **Indian Council of Medical Research (ICMR)** plays a significant role in research and surveillance related to AMR.

Addressing AMR is a multifaceted issue that involves coordination among **various government departments**, including healthcare, agriculture, and environment, as well as **collaboration with International organisations such as the WHO**.

Indian Priority Pathogen List¹¹

The Indian Priority Pathogen List (IPPL) is a significant development in the field of public health in India. It was first introduced in 2018 by the ICMR and the Department of Biotechnology (DBT). The IPPL serves as a comprehensive compilation of pathogens that pose a substantial threat to public health due to their potential to cause severe infections and the challenges they present in terms of treatment and control.

Scope and Objectives of Developing the Indian Priority Pathogen List¹¹



Creation of the IPPL was driven by the growing concern over AMR and the need for targeted surveillance and interventions.



Determine the critically significant antibiotic-resistant bacteria at the national level in India, necessitating the expedited development of innovative drugs and therapeutic interventions.



The IPPL encompasses various categories of pathogens, including bacteria, viruses, fungi, and parasites, that have been identified as critical targets for research, monitoring, and control efforts.



Strengthening of disease surveillance, enhancing diagnostic capabilities, and guiding the development of effective strategies for infection prevention and treatment.



Provide direction for the prioritization of research concerning AMR, encompassing incentives and funding allocation.

Aligned with the Global Priority Pathogen List, experts have categorized bacterial pathogens based on their species and resistance, subsequently classifying them into three priority tiers: critical, high, and medium, as illustrated in the table below:-

Table 1: Indian Pathogen Priority List¹¹

CRITICAL PRIORITY	
Enterobacteriaceae (Klebsiella pneumoniae and Escherichia coli)	Carbapenem – R Tigecycline – R Colistin – R
Non-fermenting bacteria (Acinetobacter baumannii and Pseudomonas aeruginosa)	Carbapenem – R Colistin – R
HIGH PRIORITY	
Staphylococcus aureus	MRSA, hVISA Daptomycin – NS Linezolid – R

¹¹ Department of Biotechnology. (2021). *Indian Pathogen Priority list*.

Enterococcus species	Vancomycin – R Linezolid – R Daptomycin – NS
Salmonella species (Typhoidal and Non-typhoidal)	Azithromycin – NS Third generation cephalosporins – NS Carbapenem – NS
MEDIUM PRIORITY	
Streptococcus pneumoniae	Cephalosporin – R Fluoroquinolones – R Linezolid – R
Staphylococcus, coagulase-negative	Vancomycin – R Linezolid – R
Shigella species	Third generation cephalosporins – R Azithromycin – R
Haemophilus influenzae	Third generation cephalosporin – NS Carbapenem – NS
Neisseria meningitidis	Fluoroquinolones – NS Third generation cephalosporins – NS

R: resistant; NS: non-susceptible; MRSA: methicillin resistant Staph. aureus; hVISA: heterogenous vancomycin-intermediate Staph. Aureus M

Trends in Antimicrobial Resistance

1. Antimicrobial Resistance and Humans

Antimicrobial Resistance is a significant global health threat, with its impact particularly alarming in India. In 2010, India emerged as the largest consumer of antibiotics for human health, with consumption of **12,900 Crore units**.¹²

However, despite being the largest consumer, the per capita consumption of antibiotics in India was relatively lower compared to several other countries. Specifically, India's per capita antibiotic consumption was 10.7 units per capita as compared to the US which recorded a much higher per capita antibiotic consumption of 22 units per capita in 2010.¹²

The total defined daily dosage (DDD) usage of Antimicrobials has increased to **7,976 million in 2020 from 5,411 million in 2010**.¹³

% change in total
usage 2010-2020¹³

47.4%

The DDD per capita usage of antimicrobials has risen to **5.74 in 2020 from 4.40 in 2010**.¹³

% change in per capita
usage 2010-2020¹³

30.6%

India exhibits some of the highest rates of antibiotic resistance among the bacteria responsible for common infections in both community and healthcare settings. In the year 2021, *Escherichia coli* was the most commonly isolated pathogen followed by *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Staphylococcus aureus*. These bacteria are frequently associated with infections and are of concern due to their potential for antimicrobial resistance.

2. Antimicrobial Resistance and Agriculture and Animal Farming

India accounts for 3% of global consumption of antimicrobials in *food animals** and is **fourth highest in the world**, behind **China (23%), the United States (13%) and Brazil (9%)**. The consumption of antimicrobials in the food and animal sector in India is further expected to **double by 2030**.¹²

India is the largest producer of milk and the third largest producer of fish in the world. Additionally, poultry consumption in India is projected to increase by **577% from 2000 to 2030**. Likewise, India plays a significant role in global meat, meat products, and farmed seafood production, it is anticipated that this market will grow by **312%** by the year **2030**. Given the large potential of food animal industry, antimicrobial agents are used extensively to prevent diseases in domesticated

*Food animals – These are animals that are raised for the production of food for human consumption such as eggs, meat and milk.

¹² National Centre for Disease Control. (2014). *National Action Plan, Antimicrobial Resistance*

¹³ The Centre for Disease Dynamics, Economics & Policy. (2021). *State of the World's Antibiotics*.

¹⁴ Taneja & Sharma et al. (2019) *Antibiotic Resistance in India*, IJMR.

The use of antibiotics to promote growth in food animals and poultry is a widespread practice, although the full scope of this practice remains uncertain. Critical antibiotics like tetracycline, doxycycline, and ciprofloxacin, vital for human health, are frequently employed for growth promotion in poultry. An even more alarming concern is the usage of colistin for poultry growth promotion, prophylaxis, and treatment. Research indicates the presence of antimicrobial residues in chicken meat and shrimp sold for human consumption.¹³

The burden of antimicrobial resistance in Indian livestock and food animals remains inadequately documented, with limited comprehensive studies at the national level. The absence of significant regulations on antibiotic use for non-therapeutic purposes suggests that the emergence of AMR due to antibiotic overuse in the animal sector is likely an unmeasured issue in India.¹²

3. Antimicrobial Resistance and Environment

According to UNEP, the environment and Antimicrobial Resistance are closely linked. The environment can both facilitate the spread of AMR microbes and contribute to the problem by releasing antimicrobial pollutants, which harm biodiversity and ecosystems.¹⁵

The emergence of antibiotic resistance is heavily influenced by antibiotic use, but inadequate sanitation also significantly contributes to the spread of Antibiotic-resistant bacteria and Resistance Genes (ARGs). In India, **over 50% of the population** lacks access to proper sanitation facilities for safe human waste disposal. Moreover, a substantial amount of untreated sewage is discharged into water bodies, resulting in severe contamination of rivers with antibiotic residues, antibiotic-resistant organisms, and ARGs.²

¹⁵ [United Nations Environment programme.\(2022\). Environmental Dimensions of Antimicrobial Resistance.](#)

Key Drivers of AMR



Improper Use of Antimicrobials⁶ - Misuse and overuse of antimicrobial drugs, in both human and animal healthcare by self-medicating, using antibiotics when not needed, not following the prescription leading to incomplete treatment regimen, accelerates the development of resistant bacteria.



Limited Access to Clean Water, Sanitation and Hygiene (WASH) for Both Humans and Animals⁶ - Limited access to clean water, proper sanitation facilities, and good hygiene practices increases the risk of infections and the need for antimicrobial treatment.



Infection and Disease Prevention and Control⁶ - Improper infection and disease prevention and control contributes to the transmission of drug-resistant infections. Inadequate hygiene practices, insufficient sterilisation of medical equipment, and improper use of infection control protocols create opportunities for resistant bacteria to thrive and spread.



Access to Quality, Affordable Medicines, Vaccines, and Diagnostics⁶ - The limitation of not being able to afford quality treatment, hampers effective control of infections which can lead to inadequate or inappropriate use of antimicrobials and promote the development of resistance.



Awareness and Knowledge⁶ - Insufficient awareness and knowledge about the proper use of antibiotics, the risks of AMR, and the importance of infection prevention contribute to inappropriate antibiotic use and the spread of resistance.



Enforcement of Legislation⁶ - Inadequate regulation of antibiotic sales, surveillance of antimicrobial use, and a lack of penalties for non-compliance with guidelines.

Opportunities and Challenges

Opportunities



Global Coordination

AMR has prompted the need for global coordination between the governments, institutions and international organisations. Global Action Plan for AMR, Global Research and Development priority setting for AMR, Global Antibiotic Research and Development Partnership are some of the developments towards addressing AMR.^{7,16}



Health Systems Strengthening

Investing in laboratory infrastructure, use of data analytics and machine learning, incentivising research and development for developing new vaccines, antibiotic and alternative therapies for infection prevention and control, implementing digital tracking system for tracking any misuse of antibiotics. It also includes strengthening of healthcare personnels by training and educating them about the risks and management of AMR.



Technical Leadership and International Cooperation

India has taken proactive steps to address AMR. Initiatives like BIRAC initiative promotes AMR research in the country. India also plays a leadership role in international alliances such as Global Health Security Agenda (GHSA) steering group and participating in the Global AMR Research and Development hub, demonstrating its commitment to combating AMR at a global scale.



One Health Approach

Antimicrobial Resistance Surveillance and Research Network (AMRSN) provides data on drug-resistant infections and AMR patterns in Indian hospitals, informing the development and updating of treatment guidelines. AMRSN plans to scale up to include animal, environmental and food samples, to reflect the principles of a One Health approach-based surveillance system.

¹⁶ Indian Council of Medical Research. India's Antimicrobial Resistance Surveillance & Research Initiative. Collaborations. Retrieved 20 September 2023.

¹⁷ Ranjalkar J, Chandy SJ. (2019). India's National Action Plan for AMR - An overview of the context, status, and way ahead. *J Family Med Prim Care*.

¹⁸ Ministry of Health & Family Welfare. (2016). *AMR and its Containment in India*.

¹⁹ Walia K, Sharma et al. (2019). Understanding policy dilemmas around antibiotic use in food animals & offering potential solutions. *Indian J Med Res*.



AMR Surveillance and Regulation

Closely monitoring the trends and patterns, healthcare systems of AMR across the nation is critical for progress and can help in timely implementing the targeted interventions to mitigate the spread of AMR both in humans and animals. The government of India is working on establishing country wide network for the same.

Enhancing the enforcement of existing regulations could include substantial penalties for non-compliance of the rules.



Attracting Investments

Promotion of investment opportunities in AMR activities by conducting workshops like National Stakeholder Workshop by the Department of Animal Husbandry and Dairying, and other IEC activities emphasising on the efforts required to strengthen the infrastructure to combat AMR.



Expanding Antimicrobial Stewardship

Expanding antimicrobial stewardship program across various healthcare settings, including smaller clinics and rural healthcare settings is essential. The expansion could be facilitated by tailored guidelines for different healthcare levels.



Building Awareness

Engaging local communities in AMR surveillance and stewardship through training community health workers and spreading awareness about AMR. Implementing robust initiatives to enhance access to clean water, sanitation and hygiene (WASH) can lower the infection rate and reliance on antibiotics.

Challenges

Drug-resistant pathogens can spread across borders which makes it a global problem that requires coordinated international efforts.



Cross-border Spread

AMR has a substantial economic impact affecting both productivity and healthcare expenses. In the absence of effective prevention and treatment options, treatment failures and fatalities will increase, heightening the risk associated with medical procedures including routine surgeries and infections.⁶



Economic Impact

In India, the existing studies are constrained in their scope, and it is crucial to effectively identify resistance patterns and quantify the nationwide mortality burden associated with AMR.¹⁹



Limited Research on AMR in India

Emergence of multi-drug resistant bacteria across multiple pathogens of clinical importance limits the treatment options and increases the risk of outbreaks and complications.⁶



Emergence of Multi-Drug Resistant

Due to limited awareness about infectious diseases among the general population and limited access to healthcare, people often hesitate to seek medical advice and resort to self-prescribing antibiotics without proper knowledge of the appropriate dosage and treatment duration, integrating behavioural interventions for promoting correct use of antibiotics.⁶



Limited Awareness about AMR

Uncontrolled access to antibiotics including unrestricted sales through Over-The-Counter (OTC) transactions, as well as sales without prescriptions or with invalid prescriptions is a major challenge to curb the use of antibiotics. Unregulated use of antibiotics in animals results in the accumulation of antibiotic residues in the food chain. In India, despite the presence of integrated policies for the livestock and poultry, the establishment of unified regulations with coordinated efforts is imperative to develop stringent antimicrobial usage policies.^{17,19}



Unregulated Use of Antibiotics

Cross-Cutting Themes

One Health Approach



Significance: One Health approach recognises the interconnectedness of human, animal, and environmental health making it pivotal to address public health challenges like AMR.

Policy/Programme: Government and industry collaborations are promoting a holistic approach to combat AMR. The **National Action Plan on AMR** emphasises cross-sectoral cooperation and surveillance.

Way Forward: Strengthening interdisciplinary collaboration, expanding AMR surveillance, promoting antibiotic research.

Water and Environmental Contamination



Significance: Release of antibiotics and resistant bacteria into the environment through wastewater and agricultural runoffs contributes to environmental reservoirs of resistance genes.

Policy/Programme: There are various policies and rules on waste water management by Ministry of Environment, Forest and Climate Change, **Bio-medical waste management rules** aims to improve the process of disposal of infectious bio-medical waste in a safe manner.

Way Forward: Strengthening surveillance systems, spreading awareness on responsible waste management and fostering sustainable production of drugs.

Research and Development



Significance: Development of alternative antibiotics, diagnostics and therapies are crucial to address Antimicrobial Resistance.

Policy/Programme: **National Policy for Containment of AMR** aims to conduct research in antimicrobials.

Way Forward: Encouraging public-private partnerships and fostering global collaborations for efforts on Research and Development.

Regulatory Oversight



Significance: Regulating the use of antibiotics and restricting the overuse of the same, building policies and frameworks for the surveillance of AMR is crucial to combat AMR.

Policy/Programme: **Antimicrobial Stewardship Programme Guidelines** promotes rational usage of antibiotics among the healthcare providers.

Way Forward: Implementing strict antibiotic prescription control policies, strengthening regulatory mechanisms, and ensuring compliance and enforcement.

Health Systems Strengthening



Significance: Strengthening of healthcare systems by development of facilities, training professionals for effective management of prevention of the infections and controlling antibiotic use to prevent the transmission of pathogens.

Policy/Programme: **Antimicrobial Resistance Surveillance and Research Network (AMRSN)** by ICMR have created labs and nodal centres to monitor the trends of antimicrobials and conduct research.

Way Forward: Developing infection and disease prevention and control measures, strengthening healthcare infrastructure, capacity building, awareness campaigns to enhance healthcare systems.

Public Awareness and Education



Significance: Public awareness and education are vital to ensure the responsible use of antibiotics and prevent self-medication.

Policy/Programme: **National Programme for AMR Containment and Red Line Campaign under National Response for AMR Containment** are some initiatives for public awareness about AMR.

Way Forward: Enhancing public awareness through large-scale campaigns, integrating healthcare professionals in education efforts and promoting responsible antibiotic use.

Ministries and Departments

Inter-ministerial and departmental collaboration is an important aspect of governance in India. Various ministries, departments, and organisations work together to achieve common goals and objectives.

Table 2: Ministries and Departments working towards AMR

 Ministry	 Department and Bodies	 Functions and Roles
 स्वास्थ्य एवं परिवार कल्याण मंत्रालय MINISTRY OF HEALTH AND FAMILY WELFARE Ministry of Health and Family Welfare	 National Centre for Disease Control (NCDC)	It plays a crucial role in AMR containment through surveillance, policy development, research, training, etc.
	 Indian Council of Medical Research (ICMR)	It conducts research initiatives on AMR including AMRSN to monitor and address AMR.
	 Central Drugs Standard Control Organization (CDSCO)	Responsible for drug standard control and granting licenses for special category drugs.
	 Food Safety and Standards Authority of India	Statutory body involved in food safety and regulation; responsible for licensing and registration of entities involved in food manufacturing, storage, transport or distribution of food, including dairy and dairy products.
 विज्ञान एवं प्रौद्योगिकी मंत्रालय MINISTRY OF SCIENCE AND TECHNOLOGY Ministry of Science and Technology	Department of Biotechnology (DBT)	Nodal agency supporting research and its applications in the Life Sciences, it promotes and accelerates the pace of development and large-scale use of biotechnology in the country.
	 Council of Scientific and Industrial Research (CSIR)	Provides significant technological intervention in many areas concerning societal efforts including chemicals, drugs, genomics and biotechnology.

 कृषि एवं किसान कल्याण मंत्रालय MINISTRY OF AGRICULTURE AND FARMERS WELFARE Ministry of Agriculture and Farmers Welfare	Department of Agriculture and Farmers Welfare	Regulates the use of pesticides, fertilizers, veterinary drugs and animal feed through various agencies, such as the Central Insecticides Board and Registration Committee (CIBRC), the Fertilizer Control Order (FCO), the Animal Husbandry Commissionerate, etc.
 Ministry of Environment, Forest and Climate Change	Hazardous Substances Management Division	Regulates the management of hazardous substances, waste and pollutants through various laws, rules, and guidelines.

Government Interventions

National Policy for Containment of AMR^{12, 20}

The Ministry of Health and Family Welfare launched the **National policy for containment of AMR** in 2011 acknowledging the problem of Antimicrobial Resistance.

The National Policy for Containment of AMR was developed with the objective to:

- Understand the emergence and spread of Antimicrobial Resistance and factors influencing it.
- Establish a nationwide, well-coordinated antimicrobial programme with well-defined and interlinked responsibilities and functions of different arms of the programme.
- Rationalise the usage of available antimicrobials to ensure their responsible and appropriate use.
- Reduce antibiotic selection pressures by appropriate control measures.
- Promote the discovery of newer and effective antimicrobials.
- Ensure rapid and accurate diagnosis of infections and infectious diseases by enabling targeted treatment strategies and reducing unnecessary antimicrobial use.

In November 2014, during the WHO Regional Committee meeting, member states were urged to expedite their national initiatives to enhance their capabilities for implementing the Jaipur Declaration on AMR and the South-East Asia Regional Strategy on AMR.

National Programme on AMR Containment²¹

Controlled by National Centre for Disease Control (NCDC) the Programme was launched in 2013 to establish laboratory-based AMR Surveillance System, strengthen infection control practices and to generate awareness on AMR.

The National Programme on AMR Containment was developed with the objective to:

- Establish a laboratory-based AMR surveillance system to generate quality data on Antimicrobial Resistance.
- Strengthen infection control practices and establish surveillance of healthcare-associated infections to reduce transmission.

²⁰ Directorate General of Health Services. (2011). *National Policy for Containment of AMR*.
²¹ National Centre for Disease Control. *National Programme on AMR Containment*

Under the Programme, **National AMR Surveillance Network (NARS-Net)**²¹ was developed to assess the magnitude and trends of AMR in various geographical regions of the country.

Since 2022-23, State AMR Surveillance Networks are gradually taking shape under the National Programme for AMR Containment. Six state medical colleges under NARS-Net have received support to establish these networks within their states. Nodal centres in collaboration with NCDC, are responsible for organising various activities. These include sensitisation meetings, training sessions for laboratory and data management, and review meetings for state network laboratories, which encompass government medical colleges and prominent private institutions.

The capacity building for network labs is fulfilled by including government medical colleges who are required to submit surveillance data on AMR.



National Reference Labs (NRLs): There are two NRLs - one focus on bacterial pathogens and other on fungal pathogen. The first lab is bacterial pathogens established at Centre for Bacterial diseases and Drug resistance. The second lab is for Fungal pathogens established at the fungal laboratory of Vallabhbhai Patel Chest Institute, Delhi.



Sentinel AMR Surveillance sites: Includes 40 labs across 31 states/UTs.



Figure 1 - Sentinel AMR Surveillance sites in India

National Action Plan on AMR (NAP-AMR)¹²

The NAP-AMR was launched in 2017 as a comprehensive strategy to tackle AMR in India. It outlined a multi-sectoral approach to address the issue and focuses on key areas such as surveillance, rational use of antibiotics, infection prevention and control, research and innovation, and advocacy and awareness.

The primary objective of NAP-AMR is to actively address the challenge of AMR within India and actively participate in global endeavours to combat this significant public health concern. It aims to create and reinforce governance structures and enhance the capabilities of all involved parties to diminish the repercussions of AMR in India. NAP-AMR predominantly concentrates its efforts on addressing bacterial resistance.

The NAP-AMR provides a roadmap for addressing the public health issue of AMR in India. It outlines a series of priorities and interventions aimed at effectively combating AMR which involves engagement with national disease control programmes and, at the sub-national or state level, the development of state action plans on AMR to ensure practical implementation and impact on the ground.

Three states have developed their State Action Plans in accordance with NAP-AMR

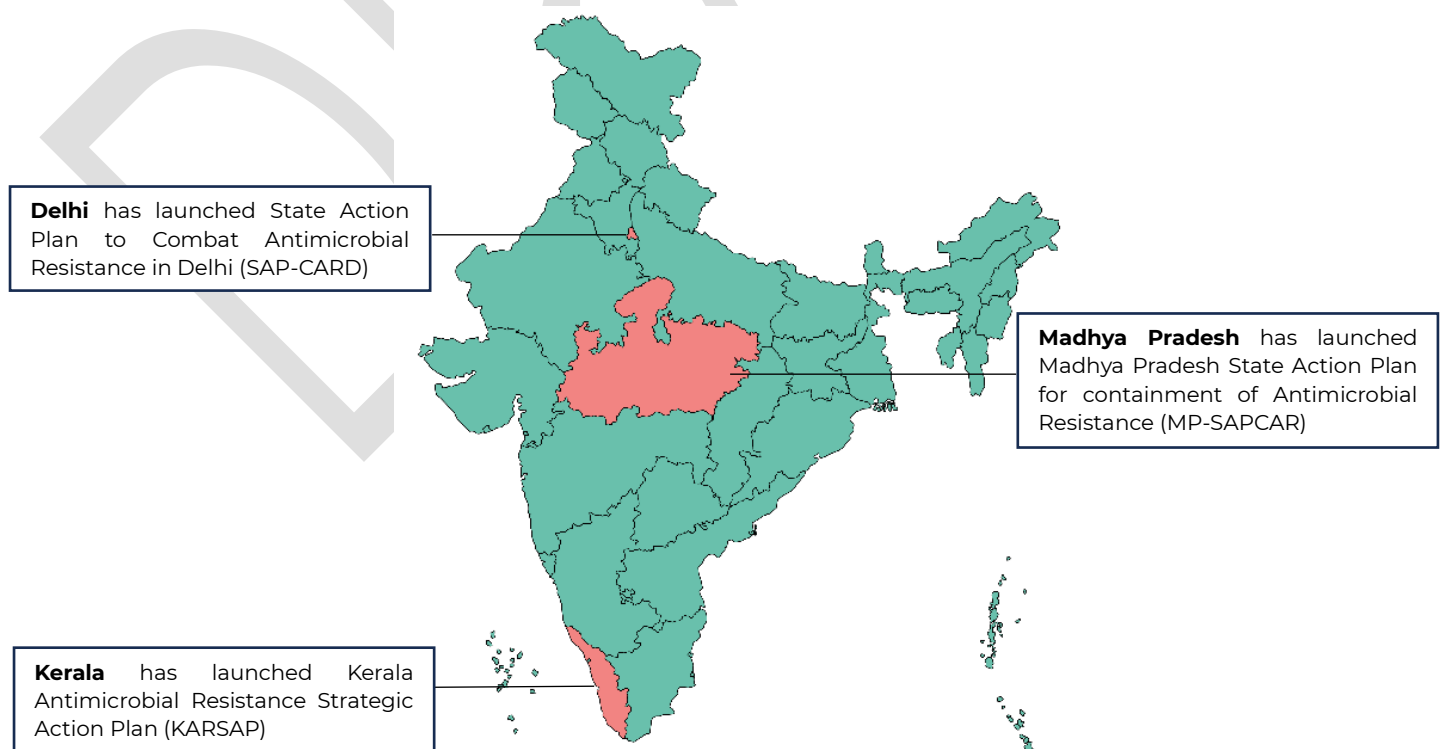


Figure 2 - States who have prepared their State Action Plans for AMR

Antimicrobial Resistance Surveillance and Research Network (AMRSN) of ICMR²²

Indian Council of Medical Research (ICMR) established the AMRSN in 2013 to generate evidence and monitor trends and patterns of drug-resistant infections in India. This network consists of 30 tertiary care hospitals, encompassing both private and government institutions.

There are four Nodal Centres and 20 Regional Laboratories across 14 states/UTs for AMR in India

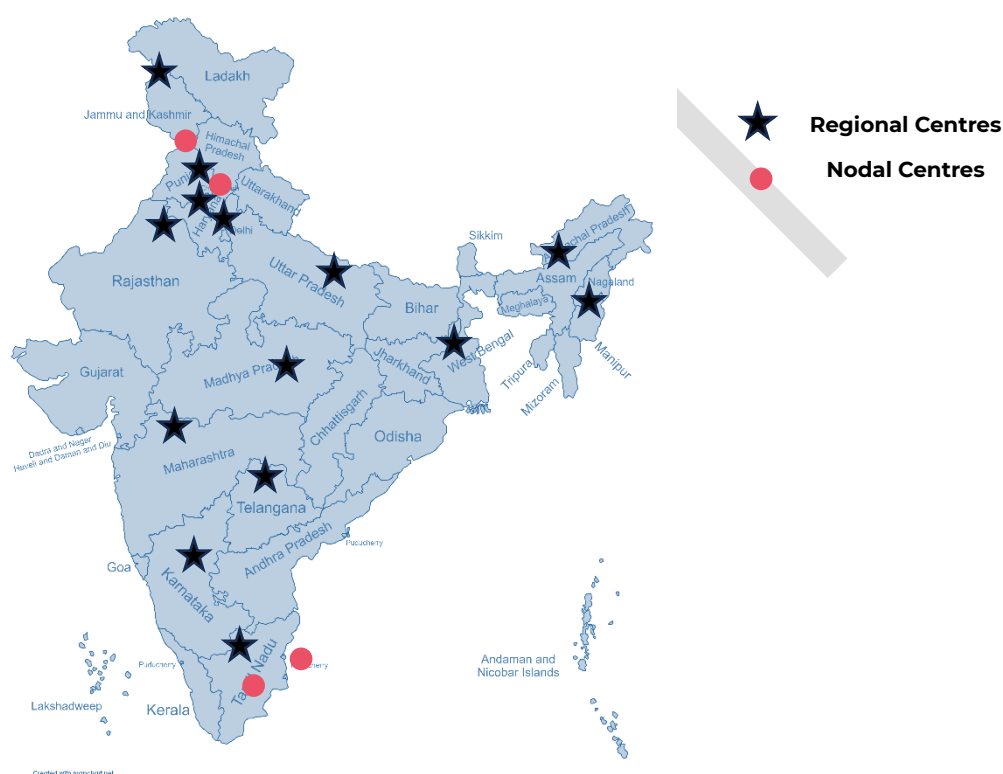


Figure 3 - Nodal Centres and Regional Laboratories of AMR in India

Antimicrobial Stewardship Programme (AMSP) Activities²³

It promotes rational use of antibiotics among healthcare providers. Sensitisation and training workshops have been conducted in various healthcare facilities nationwide, benefiting practicing clinicians.

National Centre for Disease Control (NCDC) has developed standard treatment guidelines for the rational use of antibiotics, making them accessible to clinicians across the country to enhance knowledge, improve prescribing practices, and combat Antimicrobial Resistance.

International Collaborations and Initiatives

AMR being a global issue, various global level initiatives and Alliances have been initiated to confront the issue where India has been playing a key role.

- ICMR along with Research Council of Norway (RCN) initiated a joint call for research in Antimicrobial Resistance in 2017.
- ICMR along with Federal Ministry of Education and Research (BMBF), Germany has a joint Indo-German collaboration for research on AMR.

National Response for AMR Containment - Red Line Campaign¹²

The National Policy for Containment of AMR has highlighted the significance of introducing a dedicated Schedule H1 within the Drugs and Cosmetics rules to regulate the sales of antibiotics and has established guidelines for punitive measures against entities found in violation of the regulations.

Implementation of colour-coded packaging for antibiotic strips has been proposed with the aim of restricting their usage outside tertiary care settings.

The Red Line Campaign was launched in 2016 in collaboration with Organisation of Pharmaceuticals Producers of India (OPPI) with the aim of eliminating unnecessary prescription and over-the-counter sale of antibiotics causing drug resistance for several critical diseases. It urges people **not to use medicines with red vertical line**, including antibiotics, without doctor's prescription.

Other Developments

National Guidelines for Infection Prevention and Control in Healthcare Facilities²⁴- The guidelines were released by MoHFW in January, 2020. The document serves as a crucial reference for healthcare settings, offering comprehensive guidance on measures to prevent and control infections, safeguarding patient and staff well-being. It consolidates internationally accepted Infection Prevention and Control (IPC) practices for healthcare facilities in India, aligning with national initiatives on patient safety and Antimicrobial Resistance (AMR).

Treatment Guidelines for Antimicrobial Use in Common Syndromes²⁵- The guidelines were issued by ICMR in 2019. It is designed for healthcare settings with the primary objectives of promoting the rational use of antibiotics listed on the Essential Medicines Formulary (EMF) and ensuring uniformity in the treatment of diverse infectious conditions.

22 Indian Council of Medical Research. (2021). *Annual Report*.

23 Indian Council for Medical Research. (2018). *Antimicrobial Stewardship Research Guideline*

24 National Centre for Disease Control. (2020). *National Guidelines for Infection Prevention and Control*.

25 Indian Council of Medical Research. (2017). *Treatment Guidelines for Antimicrobial Use in Common Syndromes*.

26 Director General of Health Services. (2023). *National Guidelines for Surveillance of AMR in Leprosy*

27 National Centre for Disease Control. (2018). *Guidance for developing State Action Plan for Containment of AMR*.

28 Central TB Division. (2017). *National Strategic Plan TB*

29 Central TB Division. (2023). *India TB Report*

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National Guidelines for Surveillance of AMR in Leprosy²⁶ - The guidelines were issued in 2023 by MoHFW. The guidelines were developed with the objective of detecting the primary and secondary drug resistance to anti-leprosy drugs and to standardise the reports of AMR testing.

Guidance for developing State Action Plans for Containment of Antimicrobial Resistance (SAPCAR)²⁷ - The guidelines were issued by the National Centre for Disease Control in 2018, to give a standard guideline to the states in the country for developing their own State Action Plan for the containment of AMR in accordance with National Action Plan on AMR.

Way Forward



Availability of Rapid Molecular Tests^{28,29}

- National Strategic Plan for Tuberculosis mentions the need for increasing the availability of rapid molecular tests and ensuring these diagnostic facilities are open for patients for the early detection of TB/HIV through molecular diagnostics.
- Cartridge-Based Nucleic Acid Amplification Test (CBNAAT), Line Probe Assay, and TrueNat are high-sensitivity rapid molecular tests used for the diagnosis of drug-resistant tuberculosis (TB) and the detection of rifampicin resistance/MDR-TB.
- Some of the states using the tests are, Sikkim where TRUENAT machines are accessible at Primary Healthcare Centres and Uttarakhand conducts CBNAAT and TRUENAT tests for TB diagnosis.^{25,26}



One Health Approach^{32,33}

- AMR serves as a prime example of the One Health approach. It includes collaborative efforts across diverse sectors to address challenges affecting humans, animals and the environment. AMR is intricately connected to all three of these facets.
- The Department of Global Coordination and Partnership (GCP) for AMR collaborates with the Food and Agriculture Organization (FAO), the World Organisation for Animal Health (WOAH), the United Nations (UN), along with other entities including civil society and the private sector, to lead and synchronise a global, multisectoral One Health response to AMR.



Generating Awareness Amongst Healthcare Providers¹²

Information, Education, and Communication (IEC) activities by official bodies like National Centre for Disease Control and ICMR are integral to promote awareness about AMR.



Strengthening of Surveillance System¹²

To strengthen the surveillance system, we need to make significant progress on all fronts - integration of surveillance networks, enhancement of data collection, laboratory capacity building,



Promotion of Investment for AMR activities, research and innovations¹²

Ensuring the availability of effective diagnostics, drugs, and alternative approaches for managing infectious diseases while also securing sustainable resources for AMR containment.



Strengthening Infection Control Practices¹²

Infection Control Programmes (IPC) for containment of AMR have been launched by ICMR in the form of guidelines, however, there is a need to integrate IPC programmes, both from the policy making and implementation perspective.



Strengthening India's leadership on AMR¹²

Strengthening India's leadership through international collaborations, contributing to global efforts to contain AMR, while also emphasising national and state-level collaborations to facilitate coordination among disease control programmes and stakeholders for effective action against AMR at the ground level.



Promotion of Rational Use of Antimicrobials¹²

Educating stakeholders on appropriate use and consequences of misuse, implementing antimicrobial stewardship programmes, enhancing diagnostic support, enforcing regulations, and fostering multidisciplinary collaboration for responsible prescribing and monitoring is necessary.

³⁴ National Centre for Disease Control. (2020). *State Action Plan to Combat AMR in Delhi*.

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DRAFT

1. State Action Plan to Combat AMR in Delhi (SAP-CARD)³⁴



Figure 4 - Delhi State Action Plan

Background

The Government of Delhi in alignment with the National Action Plan on Antimicrobial Resistance (NAP-AMR) and the Global Action Plan on Antimicrobial Resistance (GAP-AMR) has taken initiative to address the problem of AMR. In collaboration with multiple key stakeholders the government has created the State Action Plan to Combat AMR in Delhi.

In accordance with the action plan, two AMR committees have been established by the Government of Delhi:

Advisory Committee on AMR: The committee offers high-level policy guidance and oversight for AMR containment.

Technical Committee on AMR: The committee provides strategic and operational guidance and support for AMR containment employing a One Health approach that considers human, animal, and environmental health.

These committees aim to engage stakeholders and ensure effective oversight and monitoring of activities related to AMR containment in Delhi, fostering inter-sectoral collaboration to combat the global health challenge.

The SAPCARD assesses the present situation of AMR and identifies critical issues and challenges, and outlines key strategic objectives and interventions. It also defines key outputs and stakeholders for implementation.

There are initiatives which were previously taken by the state towards the containment of AMR including the initiatives by the Delhi Society of Promotion of Rational Use of Drugs (DSPRUD) some of which are, developing a module for the school children educating them about the overuse of antibiotics and its implications, organised National Workshops on Antimicrobial Consumption Methodology jointly with United States Agency for International Development (USAID), WHO, NCDC and Government of Delhi, conducted several training workshops training more than 8000 doctors, nurses and pharmacists building the skills for rational antibiotic use, and development of Standard Treatment Guidelines to improve the rational use of drugs including antibiotic use, patient outcomes and patient safety.

Objective

The key objectives of developing the SAPCARD are to achieve the following goals:

- Generate and improve awareness about AMR through IEC focusing on Knowledge, Attitude and Practice (KAP), behavioural studies, mass media and drive for Water, Sanitation and Hygiene (WASH).
- Strengthen the evidence through antimicrobial susceptibility testing, quality accreditation for all surveillance labs following the one health approach.
- Reduce the incidence of infection through effective Infection Prevention Control in human health and animal health and the environment.
- Regulate the use of antibiotics for humans and in animal feed by strengthening drug control departments and food safety and pollution control departments, inter-sectoral coordination and Antimicrobial Stewardship Programme (AMSP) in healthcare facilities.
- Investment promotion to conduct research for evidence of informed policy making and innovations like artificial intelligence to monitor surveillance data for containment of AMR.
- Strengthen commitment and collaboration like governance mechanisms, State collaborations and inter-sectoral mechanisms for AMR in the state.

2. Madhya Pradesh State Action Plan for Containment of Antimicrobial Resistance (MP-SAPCAR)³⁵



Figure 5- Madhya Pradesh State Action Plan

Background

The Government of Madhya Pradesh in alignment with the National Action Plan on Antimicrobial Resistance (NAP-AMR) and the Global Action Plan on Antimicrobial Resistance (GAP-AMR) has taken initiative to address the problem of AMR. Multiple key stakeholders have contributed towards the development of the Madhya Pradesh State Action Plan for Containment of Antimicrobial Resistance along with the government. Inter-sectoral Coordination and One Health approach were followed towards the development of the plan.

The previous initiatives for the containment of AMR in Madhya Pradesh include Indian Initiative for Management of Antibiotic Resistance (IIMAR, Madhya Pradesh) which promotes and creates awareness about the prudent use of antibiotic to control the spread of antibiotic resistance. Integrated Disease Surveillance Program is active in Madhya Pradesh. Efforts for IPC measures across human, animal and agriculture to limit the use of antibiotics. Madhya Pradesh Fisheries Federation conducts pre-stocking feed checks to control infections. The Government has adopted the Water, Sanitation and Hygiene (WASH) approach and developed *Maryada* guidelines with UNICEF. *Kayakalp Abhiyan* was developed which included strict compliance for waste management, infection control and hygiene promotion. Standard Treatment Guidelines was compiled by Directorate of Health Services in

2014. The Department of Public Health and Family Welfare with support from AIIMS Bhopal developed and launched State Antibiotic Policy in 2018.

Objective

The key focus areas for MP-SAPCAR are:

- Increase awareness through social media advertisements for awareness generation, focusing on role plays, FM radio platforms, KAPs, workshops, seminars, and audio-visual aids.
- Strengthen microbiology laboratories for AMR surveillance and improve surveillance through quality accreditation for all surveillance labs, NABH and NABL accreditation, availability of equipment's as per the need of SOPs.
- Effective IPC control with the formation of IPC committees in all hospitals, and assigning infection control nurses in all hospitals.
- Optimise the use of antimicrobials in human, animals and environment by stopping the sale of OTC drugs for human/animal use, banning sub-standard and spurious drugs, and no sale of antimicrobials without prescription.
- Promote investment for AMR activities by promoting of research for development of new antimicrobials, generation of funds to invest in rapid diagnostic kits, app for mobile phones for easy access to treatment guidelines.
- Research and collaboration between states and sectors, and private sector engagement.

3. Kerala Antimicrobial Resistance Strategic Action Plan (KARSAP)³⁶



Figure 6 - Kerala State Action Plan

Background

The Government of Kerala in alignment with the National Action Plan on Antimicrobial Resistance (NAP-AMR) and the Global Action Plan on Antimicrobial Resistance (GAP-AMR) has taken initiative to address the problem of AMR. The Kerala government have developed the Kerala Antimicrobial Resistance Strategic Action Plan (KARSAP) plan with the involvement of multiple key stakeholders. Inter-sectoral Coordination and One Health approach were followed towards the development of the plan.

The state has also been active in the efforts towards the containment of AMR before introducing the State Action plan and some of the initiatives included training of medical professionals with the help of Indian Medical Association, Initiation of AMR surveillance in government medical colleges and tertiary care private hospitals, application and scaling up of Infection Control Programmes for limiting antibiotic use. Rajiv Gandhi Centre for Biotechnology (RGCB), Kerala is devoted towards research of molecular biology and biotechnology.

Objective

The key focus areas for MP-SAPCAR are:

- Create awareness and understanding about AMR through IEC focusing on social media, participating In World Antibiotic Awareness Week, conducting training programs and workshops.
- Strengthen knowledge and evidence by standardising AMR surveillance, establishing AMR databases, One Health AMR surveillance system.
- Infection Prevention and control in human health and animal feed and food and the environment by reducing the healthcare associated infection rates by 20% of the existing rate by one year, achieve 50% in three years and attaining international benchmarks in five years.
- Optimise the use of antibiotics by implementing necessary regulations, surveillance of use of antibiotics by reducing OTC sales of drugs, institutional antibiogram, AMR tools to measure the consumption of antimicrobials.
- Investment promotion to conduct research and innovations for containment of AMR, define research priorities and identify potential institutes, screening of phytochemicals/ herbal extracts and development of probiotics of human origin.
- Strengthen commitment and collaboration through public-private partnerships and share information on drug resistance in state disease control programmes.

Major Pathogen Groups Under AMR

1. Enterobacterales Except Salmonella and Shigella²²

- In the year 2021, E.coli was the most commonly isolated pathogen followed by Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa, and Staphylococcus aureus. These bacteria are frequently associated with infections and are of concern due to their potential for antimicrobial resistance.
- Imipenem susceptibility of E. coli has dropped steadily from 86% in 2016 to 64% in 2021 and that of Klebsiella pneumoniae dropped steadily from 65% in 2016 to 45% in 2020 and was at 43% for the year 2021. These declining susceptibility rates indicate an increasing trend of resistance to imipenem among these bacteria.

2. Nonfermenting Gram-Negative Bacilli (NFGNB)²²

- In 2021, the recorded resistance to carbapenems in Acinetobacter baumannii was 87.5%. This high resistance rate significantly limits the availability of effective treatment options for infections caused by this bacterium.
- Pseudomonas aeruginosa, a bacterium known for causing infections in various parts of the body after surgery, such as the blood or lungs (pneumonia), has shown a consistent increase in susceptibility to all major antipseudomonal drugs.

3. Staphylococci²²

- In Staphylococcus aureus, a type of bacteria that can cause infections, there are different strains: Methicillin-Susceptible (MSSA) and methicillin-resistant (MRSA). MSSA is more sensitive to certain antibiotics, while MRSA is becoming more common and harder to treat.

4. Enterococci²²

- Among Enterococcus faecium causing bloodstream infections, 37.5% were found to be resistant to vancomycin. This indicates a concerning level of resistance to vancomycin in certain strains of enterococci, posing challenges for treatment.

5. Fungi¹⁴

- Fungi like C.auris and C.krusei, are highly resistant to fluconazole, a common antifungal drug. They show very low susceptibility rates, with only around 2.6% and 2.9% of isolates being effectively treated by fluconazole. In recent years, there has been an increase in infections caused by C.auris and C.parapsilosis.

6. Diarrheal Bacterial Pathogens²²

- Among common diarrheal pathogens norfloxacin, an antibiotic, showed poor susceptibility, except for Aeromonas and Vibrio species. Therefore, the empirical use of norfloxacin for the treatment of bacterial diarrhea is strongly discouraged.

7. Typhoidal Salmonella²²

- There have been no significant changes in the antimicrobial susceptibility pattern of Salmonella Typhi or S. Paratyphi A in India. The pattern remains consistent across all participating centres in the AMR network.

Recommended Readings

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